

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 - Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	R. V. Harsha Vardhan	Reg. No.:	192211868
Section / Batch:	CSE	Date:	17/07/2026

Code of Conduct

I R. V. Harsha Vardhan (192211868) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R Harsha

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

38	192511310	VAIBHAVA LAKSHMI K	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192524458	VASANTHAN S	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. ➤ Analyze why this occurs. ➤ Explain how a circular queue solves this problem.
40	192524083	YARRAPPAREDDY ARI SAI GOPAL REDDY	1. Consider the singly linked list: 10 → 20 → 30 → 40 → 50. Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: 1 → 3 → 5 → 7 L2: 2 → 4 → 6 → 8

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge

Signature: _____

Date: _____

Course Coordinator

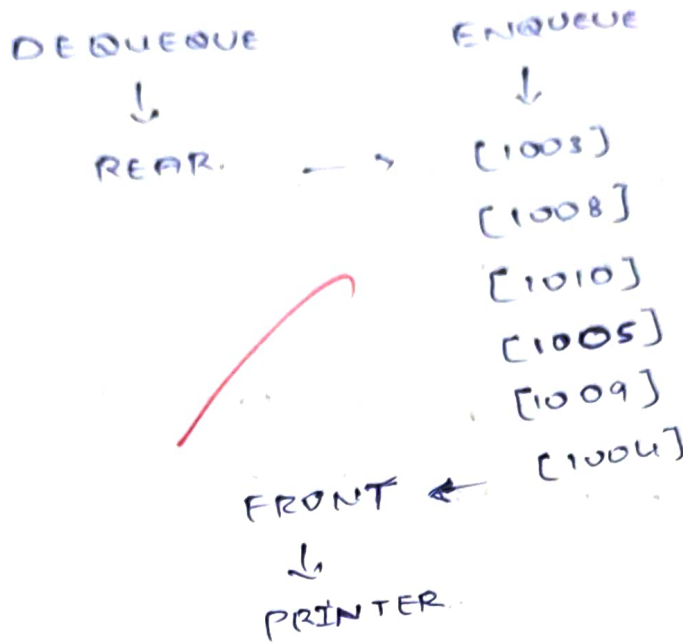
Signature: _____

Date: _____

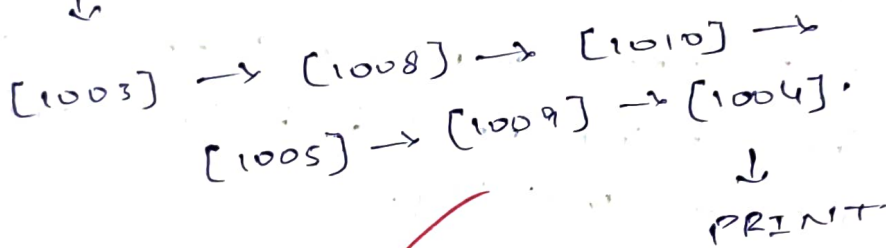
Program Director

Signature: _____

Date: _____

Printer Operations - Queue:Queue Diagram:

FRONT
REAR
↓

Printing Order:

1003
↓
1008
↓
1010
↓
1005

1009.

↓

1004.

Mechanism:

- Insert documents at the REAR of the queue
- Remove documents from the FRONT for printing.
- The first document 1003 is printed first
- The last document 1004 is printed last

Conclusion:

- A Queue is most appropriate data structure because printer requests must be handled by fairly in their arrival order
- Using FIFO prevents a newly submitted document from being printed before documents that are already waiting.

Searching for Register Number - Linear Search.

Given List:

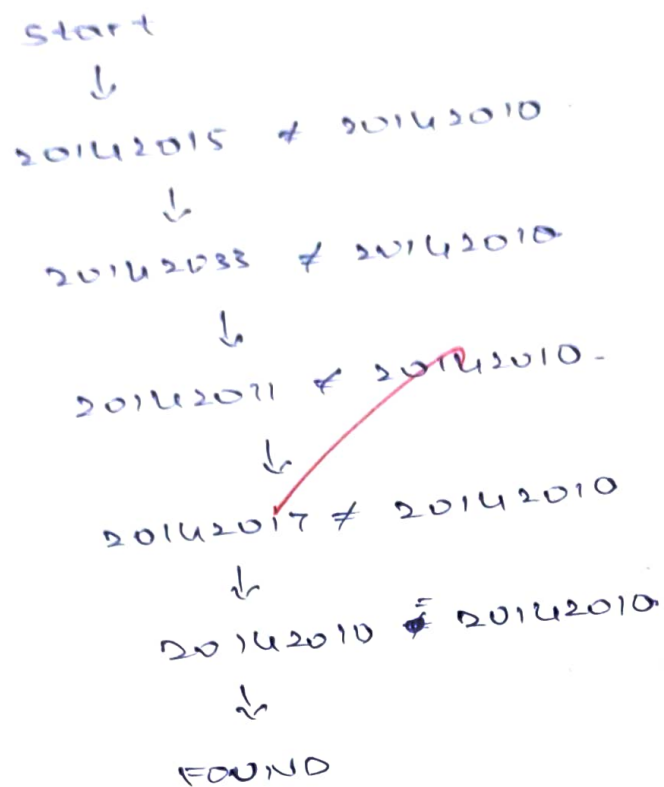
20142015
20142033
20142011
20142017
20142010
20142056
20142008.

Student's register number:
20142010.

Linear search:-

Step.	Reg No.	Comparison.
1.	20142015	≠ 20142010
2.	20142033	≠ 20142010
3.	20142011	≠ 20142010
4.	20142017	≠ 20142010
5.	20142010	→ Found.

Searching Diagram:



Conclusion:-

Linear Search is suitable because the given list is unsorted. The search starts from the first register number and continues sequentially until 20142010 is found. It is found at the 5th position, so the student is selected for placement training.

U. Jais